姓名：

学号：

学院：

上海科技大学

2015-2016 学年第一学期本科生期末考试卷

开课单位：上海科技大学生命学院

授课教师：廖军、陈佳

考试科目：生物化学 I

课程代码：

考生须知：

1. 请严格遵守考场纪律，禁止任何形式的作弊行为。
2. 参加闭卷考试的考生，除携带必要考试用具外，书籍、笔记、掌上电脑和其他电子设备等物品一律按要求放在指定位置。
3. 参加开卷考试的考生，可以携带教师指定的材料独立完成考试，但不准相互讨论，不准交换材料。

考试成绩录入表：

题目	填空题	选择题	问答题	**总分**
计分				
复核				

评卷人签名： **复核人签名：**

日期： **日期：**

编写说明：

1. 要求评卷人和复核人不能是同一人。
2. 试卷内页和答题纸编排格式由各学院和出题教师根据实际需要自定，每页须按顺序标注页码（除封面外），要求排版清晰、美观，便于在页面左侧装订。为方便印刷归档，建议使用 A4 双面印刷（学校有印刷一体机提供）。
3. 主考教师编写试卷时尽可能保证试题科学、准确、合理，如考试过程中发现试题有误，主考教师需负责现场解释，此类情况学校将作为教学评估记录的一部分。

Fill-in the blank questions

1) In a protein chain three dihedral angles are defined as _____ angles.
2) An _____ ion is a hydrogen ion associated with one or more molecules of water.
3) Disulfide bonds between cysteinyl residues can be _____ by mercaptans.
4) Supersecondary structures, also called _____, occur in many unrelated globular proteins.
5) The oxygen binding curve of hemoglobin is _____.
6) A sugar forms a cyclic-hemiacetal by reaction between an alcohol and an _____.
7) _____ base pairs can occur at the third position of the codon, as predicted by the 'wobble' hypothesis.
8) ATP is considered an "energy rich" compound because it contains two _____ bonds.
9) At the ribosome an mRNA codon must pair with the anticodon of the correct _____.
10) Phototrophs are organisms that use energy from _____ and convert it to chemical energy.
11) Compounds that contain a nitrogen base, a pentose sugar and a phosphate group are called _____.
12) Two purines found in DNA are _____.
13) Small, circular, extrachromosomal DNA molecules are _____.
14) Restriction endonucleases that make staggered cuts produce _____.
15) The linking number for underwound DNA is _____ than that for relaxed DNA.
16) Regions of DNA that influence where gene transcription starts and ends and the rate of transcription; called _____ sequences.
17) The replication _____ is a moving opening that leads the replication process.
18) In bacteria, when DNA damage is extensive, the _____ response kicks in and initiates error-prone repair.
19) _____ supercoils form ahead of the transcription bubble.
20) Hairpin loops of nascent mRNA sometimes act as _____, abruptly terminating DNA transcription to regulate gene expression.

Multiple choice questions

1. It is believed that life evolved with RNA as the genetic material, but RNA has been replaced by DNA in all current cellular life. Which feature of DNA probably accounts for this? ( )

a) DNA can direct its own replication while RNA cannot.

b) DNA is a nucleic acid while RNA is not.

c) DNA is more stable than RNA.

d) Only DNA can form the genetic material of viruses.

2. A ribozyme is: ( )

a) a class of RNA molecule that can catalyse chemical reactions.

b) a monomeric subunit of RNA.

c) a particle composed of RNA and protein that is involved in the synthesis of proteins.

d) a protein enzyme that catalyses the synthesis of RNA.

3. ATP (adenosine triphosphate) can be termed the universal energy currency in life because it: ( )

a) enables cells to disobey the first law of thermodynamics.

b) enables cells to disobey the second law of thermodynamics.

c) releases energy as heat when it undergoes hydrolysis.

d) traps energy released from the oxidation of different classes of food molecules.

4. Which term below best defines the 'quaternary structure' of a protein? ( )

a) The arrangement of two or more polypeptide subunits into a single functional complex

b) The folding of the polypeptide backbone in three-dimensional space.

c) The interaction of amino acid side chains.

d) The sequence of amino acids in a polypeptide chain.

5. Which of the following most accuratelytely charged amino acid side chains. ( )

a) Through ionic bonds operating between oppositely charged amino acid side chains.

b) Through covalent bonds joining different parts of the peptide backbone.

c) Through hydrogen bonds between different amino acid side chains.

d) Through hydrogen bonds joining different parts of the peptide backbone.

6. Which of the following amino acids is mostly likely to disrupt an alpha helix? ( )

a) Proline.

b) Leucine.

c) Glycine.

d) Valine.

7. How many different transfer RNA molecules are present in a cell (not including those present in the mitochondria)? ( )

a) 64

b) 61

c) 20

d) More than 20, less than 61.

8. Proteasomes destroy proteins by unfolding them and chopping them into small fragments. What kinds of proteins are recognized by proteasomes? ( )

a) Proteasomes recognise ubiquitinylated proteins.

b) Proteasomes recognise denatured proteins.

c) Proteasomes recognise proteins with basic or aromatic amino acids at the N-terminus.

d) Proteasomes recognise viral proteins.

9. Which of the following statements about the use of mass spectrometry in protein investigation are correct? Please select all that apply. ( ).

a) Mass spectrometry is used to rapidly determine the molecular weight of a protein.

b) Mass spectrometry can be used to determine the sequence of peptides or proteins.

c) Mass spectrometry can be used to investigate post-translational modifications.

d) Mass spectrometry may involves separating ionic fragments on a gel.

10. Which of the following statements about protein separation by gel filtration are correct? Please select all that apply. ( )

a) When a mixture of proteins is separated by gel filtration, the smallest molecular weight protein is eluted first.

b) When a mixture of proteins is separated by gel filtration, the largest molecular weight protein is eluted first.

c) When a mixture of proteins is separated by gel filtration, the smallest molecular weight protein is eluted last.

d) When a mixture of proteins is separated by gel filtration, the smallest molecular weight proteins flow around the beads.

11. Which of the following is true for DNA? ( )

a) Phosphate groups project toward the middle of the double helix.

b) Deoxyribose units are connected by 3',5'-phosphoanhydride bonds.

c) The 5' end of both strands are at the same end of the double helix.

d) G/C pairs share three hydrogen bonds.

12. In the DNA sequencing by the Sanger (dideoxy) method: ( )

a) the template strand of DNA is radioactive.

b) enzymes are used to cut the DNA into small pieces, which are then separated by electrophoresis.

c) ddTTP is added to each of four reaction mixtures prior to synthesis of complementary strands.

d) the role of ddATP is to occasionally terminate synthesis of DNA where dT occurs in the template strands.

13. The phosphodiester bonds that link adjacent nucleotides in DNA: ( )

a) are positively charged.

b) join the 3' hydroxyl of one nucleotide to the 5' phosphate of the next.

c) always link A with T and C with G.

d) are positively charged and always link A with T and C with G.

14. DNA supercoiling: ( )

a) results in compaction of the DNA structure.

b) makes the DNA molecule relaxed.

c) occurs only in viruses.

d) usually results from overwinding.

15. Human chromosomes are extremely large, complex structures. Which of the following statements correctly describes the organization of human chromosomes? ( )

a) All the genetic information of the cell is encoded in the nuclear, chromosomal DNA.

b) Genes for histone and ribosomal RNAs are structural genes.

c) Most of the chromosomal DNA codes for protein.

d) All the above statements are correct.

16. Consider the mRNA sequence: (5') AAUGCAGCUUUAGCA (3'). The sequence of the *coding* strand of DNA is: ( )

a) (5') ACGATTTCGACGTAA (3').

b) (3') TTACGTCGAAATCGT (5').

c) (5') AATGCAGCTTTAGCA (3').

d) (3') AATGCAGCTTTAGCA (5').

17. What is found *within* the transcription bubble? ( )

a) Positive supercoils in the DNA.

b) Negative supercoils in the DNA.

c) An intact DNA duplex.

d) A short DNA-RNA hybrid.

18. Prokaryotic *and* eukaryotic promoters: ( )

a) contain AT-rich regions.

b) interact with transcription factors.

c) are influenced by base sequences thousands of base pairs away, which increase their activity.

d) have all of the above properties.

19. CRP has been found to contain a typical helix-turn-helix motif. Which of the following correctly describes the interactions between CRP and the *lac* operon. ( )

a) Covalent bonds occur between the DNA helix and CRP helices.

b) Nonpolar interactions and hydrogen bonds contribute to the specificity of the CRP-DNA interactions.

c) The double helix of the operon DNA interacts with the double helix of the CRP DNA.

d) The base sequence of the DNA has no influence on the binding of CRP.

20. Which of the following statements about enhancer elements is correct. ( )

a) They can be located on either side of the gene they activate.

b) They will not function unless located at some distance from the gene they control.

c) They bind transcription factors.

d) An enhancer element activating a specific gene is not functional in all cell types.

Cumulative

1. Glucose-6-phosphate is formed from glucose in an ATP-utilizing reaction.

a) Using the following information, calculate the overall ΔG^{0}' for this reaction.

$$\text{Glucose} + PO_4 \rightleftharpoons \text{Glucose-6-phosphate} + H_2O \qquad \Delta G^{0} = +11.6 \text{kJ/mol}$$

$$ATP + H_2O \rightleftharpoons ADP + PO_4 \qquad \Delta G^{0} = -32.5 \text{ kJ/mol}$$

b) Write out the overall reaction.

c) In which direction will the reaction proceed?

d) Given your answer in a), is the overall reaction **spontaneous** or **not spontaneous**?

2. Soy sauce is one common ingredient in Philippine cooking. It is made by fermenting a mixture of mashed soybeans (mostly protein) and flour (mostly starch), followed by brining (soaking the fermented mixture in salt water). The brining liquid becomes the soy sauce.

a) To what class of macromolecules does starch belong?

b) What are the structural features of the polymers in starch? Choose all that apply: b1) Linear; b2) Branched; b3) Helical; b4) Extended; b5) Rigid; b6) Hydrated; b7) Anhydrous; b8) Fibrous

c) What are the monomer units of starch? Be as specific as possible

d) Name the linkages that connect these monomers in starch. Be as specific as possible

e) Briefly state the most likely purpose of adding flour to the soybeans (10 words or fewer.)

3. You wish to determine the relative degree of sequence similarity for DNA from different species of Galapagos finches (referred to as species A, B and C). You make hybrid duplexes of the DNA between A and B, A and C, and B and C. You measure the increase in the

absorption of UV light with increasing temperature for each of the duplex solutions and obtain the following data:

Hybrid Duplex	t_m
A + B	80 °C
A + C	87 °C
B + C	83 °C

a) Which two species have a greater degree of similarity in their DNA sequences? Why?

b) What factors should be taken into account when interpreting these data?

4. The *Homo sapiens* genome has approximately 3×10^9 base pairs;

a) if an "average" protein contains 400 amino acids, what is the maximum number of proteins that can be encoded by the human genome?

b) Analysis of the sequence of the human genome suggests that there are 30,000-40,000 protein-coding genes, and it is estimated that human cells make ten times this number of proteins due to differential splicing, etc. Using the information from part (a), what percentage of the total genome is actually used to encode proteins? How would you explain the answers you obtained for (a) and (b)?

c) The calculation made in part (b) above is an underestimation of the percentage of the genome used to encode proteins. Why?